

The Photonic Universe Hypothesis

A proposal that the universe is made of light, that matter is light folded into a crystal, and that every star and planet carries a seed of the beginning at its centre.

Brian Martell — Independent Researcher

Start here

New to this? Read in this order:

1. [Plain-language overview](#) — what the idea is, no equations, about ten minutes.
2. [Executive summary](#) — the claims, and what would disprove them.
3. [Glossary](#) — every technical term, defined in ordinary words.

Looking for the technical work? Go to [the archive](#).

The idea, in plain words

Everything you are made of was once light.

That much is ordinary physics. What this project proposes is a specific answer to *how*: that light does not simply become matter, but **folds** into a structure — a lattice, something like a crystal, but built from the geometry of light itself. Matter is what a fold looks like from the outside. A particle is a knot in the weave.

Three consequences follow, and they are the reason this is worth testing.

Gravity is not a force. It is what happens when the lattice packs more tightly in one place than another. Things fall the way water finds a drain — not pulled, but following the shape of what surrounds them.

Nothing ever becomes infinite. Where standard physics predicts a singularity at the heart of a black hole — a point of infinite density where the mathematics stops working — this framework predicts something finite instead: a **Planck core**, unimaginably dense but bounded, because the lattice can only fold so far.

Every massive body has one. Not only black holes. Stars and planets — including this one — are proposed to have formed *around* tiny primordial cores left over from the beginning of the universe. If that is right, the Earth has a seed of the Big Bang at its centre, and has had one the whole time.

That last claim is the one that can be checked. Checking it is the point.

Why this is offered as science and not as a story

Anyone can propose that the universe is made of light. What makes a proposal scientific is that it can be **proven wrong**, and that its author says in advance exactly what would do it.

This archive does that, on the record and before the data:

- **Predictions are registered before they are tested.** The conditions that would falsify a claim are written down and timestamped *first*, so the verdict cannot be adjusted afterwards to fit the result.
- **Errors are corrected in public.** Where earlier papers are wrong, later papers say so by number and section, and the correction stays in the record permanently. Several papers here exist for no other reason.
- **Predictions are live now.** Among them: a required relationship between the sizes of cosmic voids and the spacing of the filaments between them, testable against current sky surveys; a predicted flattening in the mass–radius relation of stars; and a floor on the mass of the smallest primordial cores, which existing observations already constrain.

The governing principle of the whole project, stated plainly:

Truth comes with mistakes — as long as we catch them early.

The archive

The complete body of work — over two hundred papers, each with a registered DOI and a priority date — is archived on Zenodo and Academia.edu.

- **Zenodo** — the authoritative record. Every paper, every correction, permanently citable.
- **This repository** — selected papers, summaries and reading guides.

The papers build in sequence and cite one another by number. The technical core develops the universe as a condensate on the **E8 lattice** — an exceptional structure in eight

dimensions, discovered by mathematicians long before anyone proposed a use for it — with gravity emerging as entropic packing, and the Planck core as a non-singular ground state.

If you want the mathematics, start with the archive. If you want the argument, start above.

Method, stated openly

The ideas in this project are mine. The mathematics and physics were developed in collaboration with AI systems — principally Claude (Anthropic), credited in the byline of every paper.

That collaboration is worth naming rather than hiding, and worth describing precisely rather than vaguely:

- **The questions, the framework and the physical intuition are mine**, as are the decisions about what to pursue and what to discard.
- **The formal derivations were built jointly**, checked line by line, and re-derived rather than recalled.
- **Other AI systems are treated as leads to verify, never as sources**. Nothing enters the archive because a model asserted it.
- **Where the collaboration got something wrong, the correction is filed** under the same names as the original.

I am a retired detective, not a physicist. Thirty-three years of investigative work taught me one thing that transfers directly: follow the evidence, write down what would prove you wrong, and never fall in love with a theory you have not tried to break.

Why I am doing this

Because the question of where we came from belongs to everyone, and too much of the answer sits behind language most people were never given.

If this framework is right, it will be tested and it will survive. If it is wrong, the record will show exactly where and exactly when — and that is a contribution too. Either outcome is better than the question going unasked.

Contact

Corrections, challenges and questions are welcome — particularly from anyone who wants to test a prediction. Open an [issue](#), or get in touch through the archive.

This work is offered openly. Please cite the DOI of the specific paper rather than this page.